

Adaptive Multi-Signal Systematic Trading Strategy for S&P 500 Stocks¹

MSc Financial Technology

¹ Word Count: 3281

Project Specification

Project Description

Objectives of Research: This project constructs an adaptive multi-signal systematic trading strategy to improve the return predictive ability and portfolio robustness by integrating multiple α signals and a dynamic risk filtering mechanism.

Scope of Research: The research objects of this project are partial S&P 500 stocks. The stock data is collected from the Bloomberg Terminal (includes price, fundamental, and implied volatility data), and the news data comes from the FNSPID dataset (Dong et al., 2024). All data covers the period from 2012 to 2020.

Technical Approach: In feature engineering, multi-factor signals are constructed through rolling windows. In the model building process, Lasso, XGBoost, and VSN+LSTM models are used for return prediction, and an AdaBoost meta-model is also applied for transaction filtering. Risk management methods include equal weight allocation, volatility, and risk parity.

Evaluation Method: The backtest is conducted on the out-of-sample data from 2017 to 2020, and the evaluation indicators include cumulative return, Sharpe ratio and maximum drawdown, without considering any transaction costs (transaction fees and slippage points).

Research Contribution

Data Aspect Innovation: Integrating price, fundamentals, implied volatility of options, and text features based on sentiment.

Methodological Integration: This project integrates rolling window multi-factor screening, Lasso and VSN+LSTM prediction, and AdaBoost meta-model for risk filtering.

Text Processing Innovation: Applying embedding and sentiment extraction based on FinBERT and aggregation through sentiment probability weighting.

Signals Construction Innovation: Based on the existing trading strategy theories, this project expands some innovative signals with high predictive capability.

Attention: All codes, data sets, project details can be found in the links in [Appendix e](#).

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² Due to the need to describe many feature mining processes and machine learning methods, the word count exceeds 3,000 words

1. Introduction

In recent years, systematic trading strategies have attracted much attention in financial markets due to their reliable performance in stock investment. However, single signal-based strategies often work badly when the market regime changes. For this reason, this project is aimed to develop an adaptive multi-signal model that integrates technical indicators, fundamental valuation factors, and sentiment-based text features from financial news. Lasso, XGBoost and VSN+LSTM are used in the prediction model, and the AdaBoost meta-model is introduced to screen trading signals according to the market state to improve robustness and reduce drawdown. The strategy also evaluates three position management methods: equal weight allocation, volatility parity and risk parity.

2. Data Preprocessing

[Appendix a](#) provides the structure of the stock data. For the text data, the news title is used as the main input. Beside this, the news data is found that a single stock may have multiple news on the same day through news data structure analysis ([Appendix b](#)). Therefore, it needs a mechanism to aggregate daily news.

In the missing values handling process, stocks with missing prices or feature missing rates over 20% were deleted, and the rest missing value were filled with the past 5-day average ([Appendix c](#)). Among the outliers, the abnormal implied forward price was directly replaced with the price when gap between the implied forward price and the price was significant ([Appendix d](#)).

3. Feature Engineering

Feature Evaluation Method

The evaluation only uses samples from 2012-01-01 to 2016-12-31 (the training set of the first rolling window). The correlation between the features and excess forward returns (1, 5, 20, 40 days) is tested by pooled OLS regression (all tickers' train data are regressed together), and the t-value of the coefficient β is calculated to assess statistical significance.

The forward return is calculated as:

$$ForwardReturn_{i,t}^{(h)} = \frac{P_{i,t+h}}{P_{i,t}} - 1$$

The excess forward return is:

$$ExcessReturn_{i,t}^{(h)} = ForwardReturn_{i,t}^{(h)} - h \cdot RF_{daily,t}$$

Pooled OLS regression:

$$Return_{t+h} = \alpha + \beta \cdot Signal_t + \varepsilon_t$$

The t-value of β is used to test the null hypothesis $H_0: \beta = 0$. When $|T| > 2$, the predictive power of signal is statistically significant.

In addition, all signals are constructed based on a rolling window, and different window lengths w may affect the predictive stability and significance of signals. To determine the optimal window w for each feature, the absolute t-value $|t_{w,h}|$ is calculated over multiple forward excess returns H . Subsequently, a scoring mechanism combined the score of the weakest significance signal and the average score among all forward return is applied to determine the best window:

Minimum absolute t-value:

$$T_{min}(w) = \min_{h \in H} |t_{w,h}|$$

Average absolute t-value:

$$T_{avg}(w) = \frac{1}{|H|} \sum_{h \in H} |t_{w,h}|$$

Composite rank score:

$$Score(w) = R_{min}(T_{min}(w)) + R_{avg}T_{avg}(w)$$

Score(w) is a rank-based aggregated score that considers two aspects at same time:

1. $R_{min}(w)$ - Ranking score of weakest significance among all forecast horizons (smaller is worse)
2. $R_{avg}(w)$ - Score of average ranking of all forecast horizon significance (larger is better overall).

Then, the two ranking scores are added together to obtain $Score(w)$, where a higher score means that the window can not only maintain a stable minimum t-value among all forward returns but also works well in the overall average performance. Therefore, those constructed signals are more stable.

If the scores are the same, the window is selected with the maximum average t-value.

a. Carry Signal (Kojien et al., 2016)

Daily Carry signals measure relative differences between the current stock price S_t and market implicit one-month forward price F_t^{1M} : $\text{Carry}_t = \frac{S_t - F_t^{1M}}{F_t^{1M}}$

When a stock's current price is higher than its implied forward price, the market expects the price to fall in the future. Conversely, it is expected to appreciate. Bloomberg's option pricing model (Black-Scholes /Merton) gives the implied forward price, which reflects the market's price expectations.

To reduce noise, the carry signal is also smoothed with exponentially weighted moving average (EWMA) of different Windows (window lengths: 5, 20, 120, and 250 trading days).

Result Analysis: As shown in [Figure 1](#), under different EWMA smoothing windows and forward excess return, the overall absolute t-value of the Carry signal is low, which is inadequate in significance. Therefore, this signal was not considered in the subsequent model construction.

Best EWMA Carry Signal Span: 5

- Min |t|: 0.24

- Mean |t|: 0.35

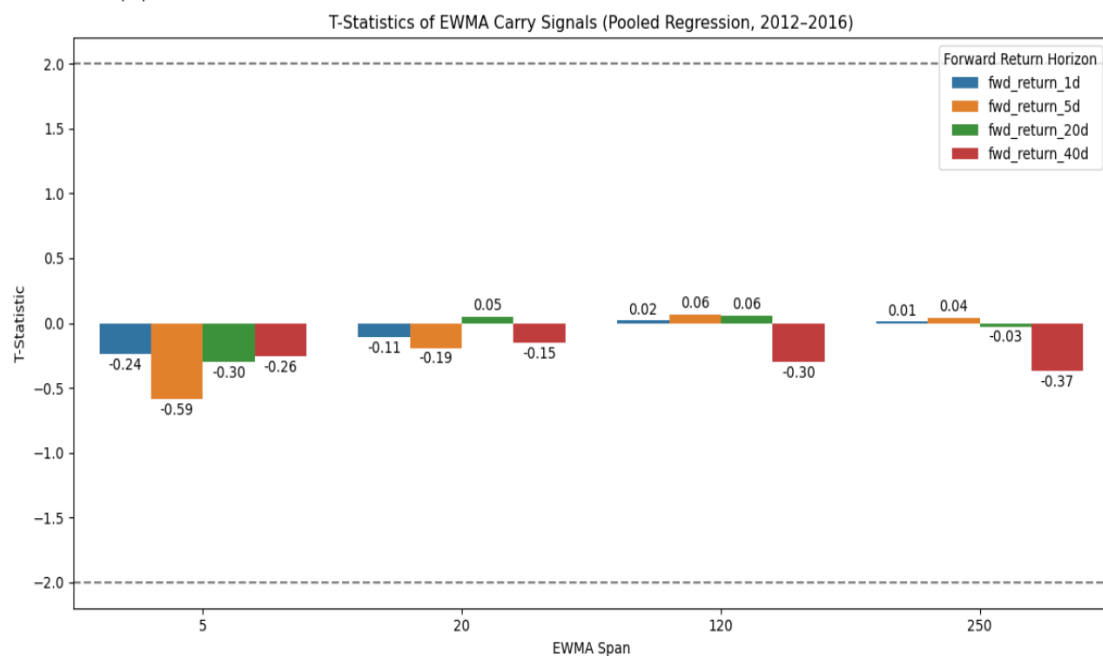


Figure 1 Carry signal OLS result

b. Volatility Risk Premium Signal (VRP)

VRP measures the difference between historical realized volatility and implied volatility:

$$VRP_t = \frac{\sigma_t^{\text{real}} - \sigma_t^{\text{impl}}}{\sigma_t^{\text{impl}}}$$

σ_t^{impl} : Implied volatility of options (IVOL_DELTA), it reflects the market's expectations of future volatility

σ_t^{real} : Historical realized volatility, it can be estimated by standard deviation (Std) or mean true volatility (ATR)

Why use ATR?

Average True Range considers intraday fluctuations and overnight gaps, which is closer to the actual risk level and may enhance the effectiveness of the signal (Wilder, 1978):

$$\text{ATR: } \tau_t = \max\{High_t - Low_t, |High_t - Close_{t-1}|, |Low_t - Close_{t-1}|\}$$

$$\text{ATR annualized realized volatility: } \sigma_t^{\text{ATR}} = \frac{EMA_n(\tau_t)}{Close_t} \times \sqrt{252} \times 100$$

Economic Meaning: When VRP is positive (historical volatility is higher than the market's expected volatility), it means that the market may underestimate future risks. Therefore, investors will require a higher risk premium, which may depress the current price. Conversely, when the VRP is negative, the current price may be pushed up.

Result Analysis (Figure 2):

Although ATR can more accurately reflect the actual volatility of intraday and overnight gaps, its predictive significance is generally lower than the standard-deviation-based VRP signals in the OLS regression test.

From the standard deviation VRP aspect, the 250-day volatility window shows a higher absolute t-value over all forward returns, indicating a more stable forecasting ability in historical data. Therefore, the Std and 250-day window are finally selected as the construction method of the VRP signal.

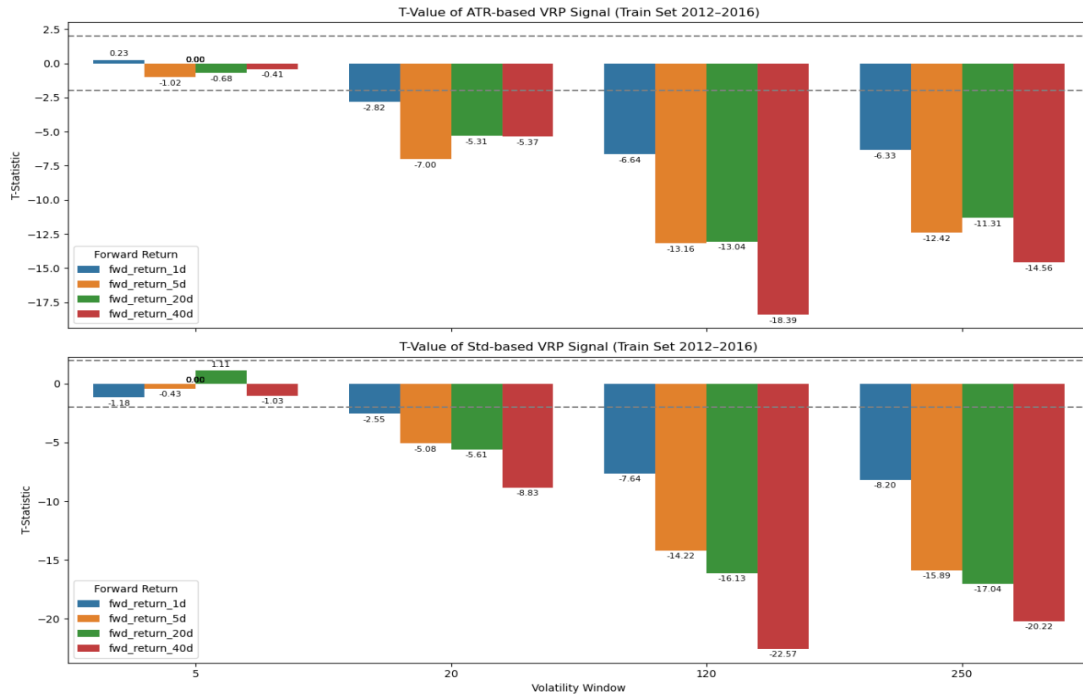


Figure 2 ATR vs STD within different EWMA windows

c. Value Signal

Signal Construction:

Firstly, given today's valuation indices V_t (Price-to-Earnings Ratio, Price-to-Book Ratio, Price-to-Sales Ratio), calculate their quantile in the historical distribution within the window length n (20, 60, 120, 250 days): $q_t = \frac{|\{i \in \{1, \dots, n-1\} : V_{t-i} \leq V_t\}|}{n}$

Secondly, standardize quantile q_t to the interval $[-1, 1]$ to obtain the final signal: $S_t = 1 - 2q_t$. The higher S_t (close to 1) means the price may undervalued, and the lower S_t (close to -1) means the price may overvalued.

Economic Meaning: When S_t approaches +1, it indicates undervaluation and potential for price appreciation; when it approaches -1, it indicates overvaluation and the risk of a price pullback.

Result Analysis (Figure 3): The t-value directions of the three valuation signals (PE, PB, PS) are opposite to the mean reversion assumption, showing a momentum effect. This may stem from investors' behavioral bias, making the price changes continuous. Among all the windows, the t-value of the 60-day window is the most statistically significant. Therefore, 60 days are selected as the best window.

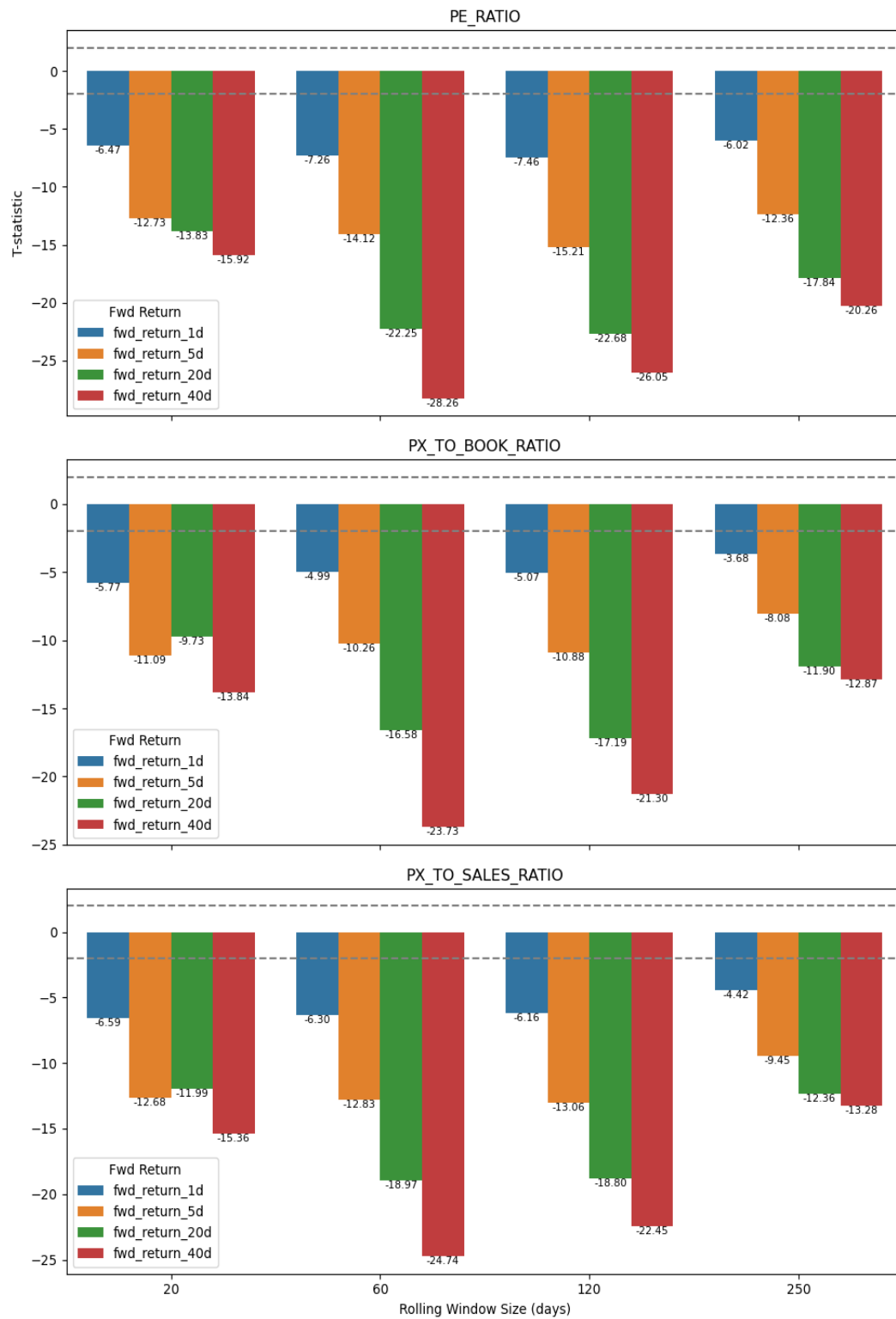


Figure 3 OLS Result for Value Signal

d. Classic Breakout and Continuous Breakout Signal - Brock et al. (1992) & Moskowitz et al. (2012)

At each time t , the signal b_t is updated according to the following rule:

$$b_t = \begin{cases} 1, & \text{if } P_t = \max(P_{t-\mu+1}, \dots, P_t) \\ -1, & \text{if } P_t = \min(P_{t-\mu+1}, \dots, P_t) \\ 0, & \text{if } P_t = \max(P_{t-\lambda+1}, \dots, P_t) \text{ or } P_t = \min(P_{t-\lambda+1}, \dots, P_t) \\ b_{t-1}, & \text{otherwise} \end{cases}$$

Where P_t is the price at time t .

Figure 4 Classic Breakout Signal Definition

Classic Breakout Definition (Figure 4 above): A buy signal is issued when the price reaches a long-term (μ) high and a sell signal is issued when a long-term (μ) low is hit. If the price reaches a short-term (λ) high or low, the signal is reset to zero to reduce excessive trading; otherwise, the previous day's signal is continued.

The signal s_t at time t is updated as:

$$s_t = \delta \cdot s_{t-1} + 1(P_t > \max(P_{t-\mu}, \dots, P_{t-1})) - 1(P_t < \min(P_{t-\mu}, \dots, P_{t-1})) + 1(P_t > \max(P_{t-\lambda}, \dots, P_{t-1})) - 1(P_t < \min(P_{t-\lambda}, \dots, P_{t-1}))$$

Where P_t is the price at time t , and $1(\cdot)$ is the indicator function.

Figure 5 Continuous Breakout Definition

Continuous Breakout Definition (Figure 5 above): This signal combines long-term and short-term breakouts and use an attenuation factor (Decay = 0.1, 0.3, 0.5, 0.7 and 0.9), allowing the signal to weaken when there is no breakout gradually. If the price breaks through the highest price of the long-term or short-term window, it strengthens the long position signal; if the lowest price is broken, strengthen the short-selling signal. This mechanism can capture both long-term and short-term momentum changes simultaneously.

Economic Meaning: Classic Breakout captures trends by relying on long-term and short-term price breaks, and short-term extreme value reset signals to prevent excessive trading. Continuous Breakout adopts attenuation factors and smoothing mechanisms to retain trend memory and reduce noise.

Results Analysis: In terms of window selection, Classic Breakout (Figure 6) shows that its significance gradually weakens when the long-term window μ increases, while it performs best when the short-term window $\lambda = 20$. Therefore, the best combination is $(\mu, \lambda) = (60, 20)$. Continuous Breakout (Figure 7) shows that the higher the attenuation coefficient δ (the slower the signal attenuation), the stronger significance, and the best effect is when $(\mu, \lambda, \delta) = (60, 5, 0.9)$. However, the t-values of both signals are negative, contrary to the trend following expectation. This may be caused by the pullback or reversal which often accompanies a price breakout, reflecting the mean-reverting of the market price.

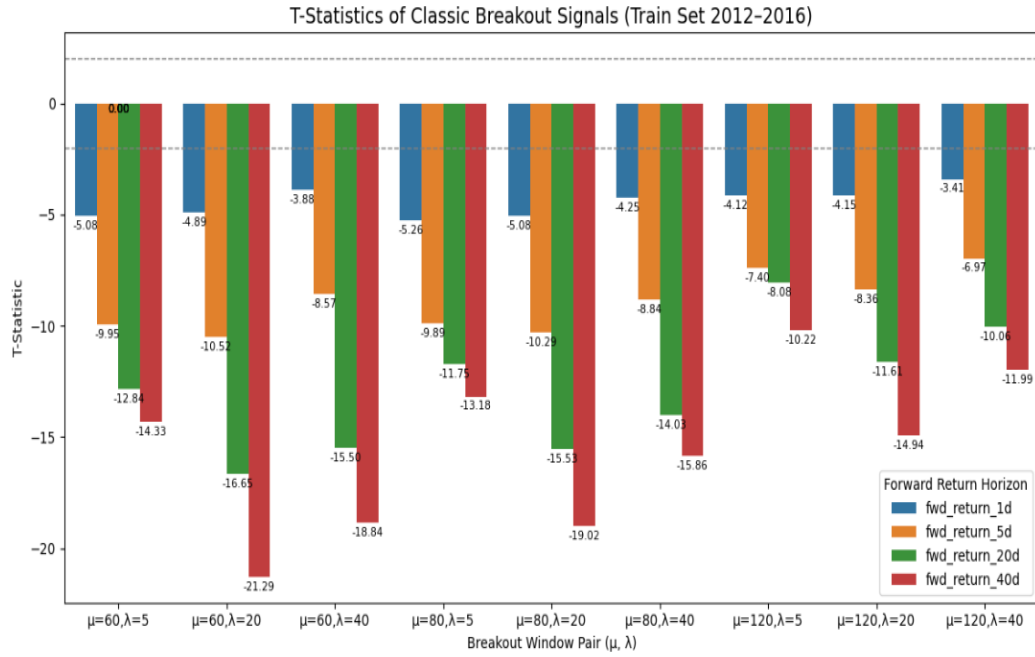


Figure 6 OLS Result of Classic Breakout

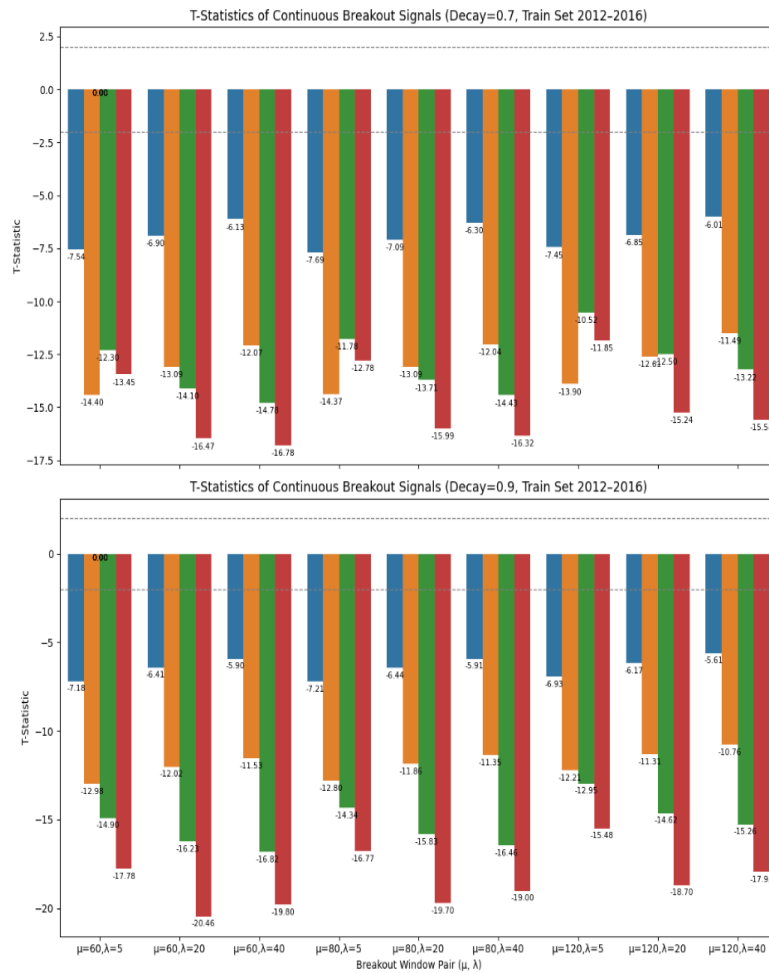


Figure 7 OLS Result of Continuous Breakout

e. Diversified Momentum Signal (Hurst et al., 2017)

The core idea of the HOP (Hurst, Ooi, Pedersen) momentum signal is to construct a multi-scale momentum indicator based on the excess returns adjusted for volatility, rather than directly using price changes.

Firstly, use the Exponential Weighted Moving Average (EWMA) of Average True Range (ATR, mechanism shown in VPR) to estimate the short-term volatility; Then divide the excess return by the volatility to obtain the standardized excess return: $s_t = \frac{r_t^{\text{excess}}}{\sigma_t}$. This step measures the unit risk return, which allow the basic model to compare the signals among different stocks.

Finally, accumulate these standardized returns over different windows (20, 40, 60, ..., 240 days), and select two windows with the largest t-value under each forward return horizon. After deduplication, obtain the final diversified momentum signals (retain 20, 60, 80, 200, 220 days momentum).

Momentum Signal Formula: $\text{Momentum}_t^{(w)} = \sum_{j=1}^w s_{t-j}$

Results Analysis: The short-term window (20-60 days) shows a significant negative t-value, which indicates short-term momentum signals may demonstrate a mean reversion of the stock price. The long-term window (100-240 days) shows a significant positive value, especially the t-value of 200-240 days, which is close to 9, demonstrating the continuity of a strong trend ([Figure 8](#)).

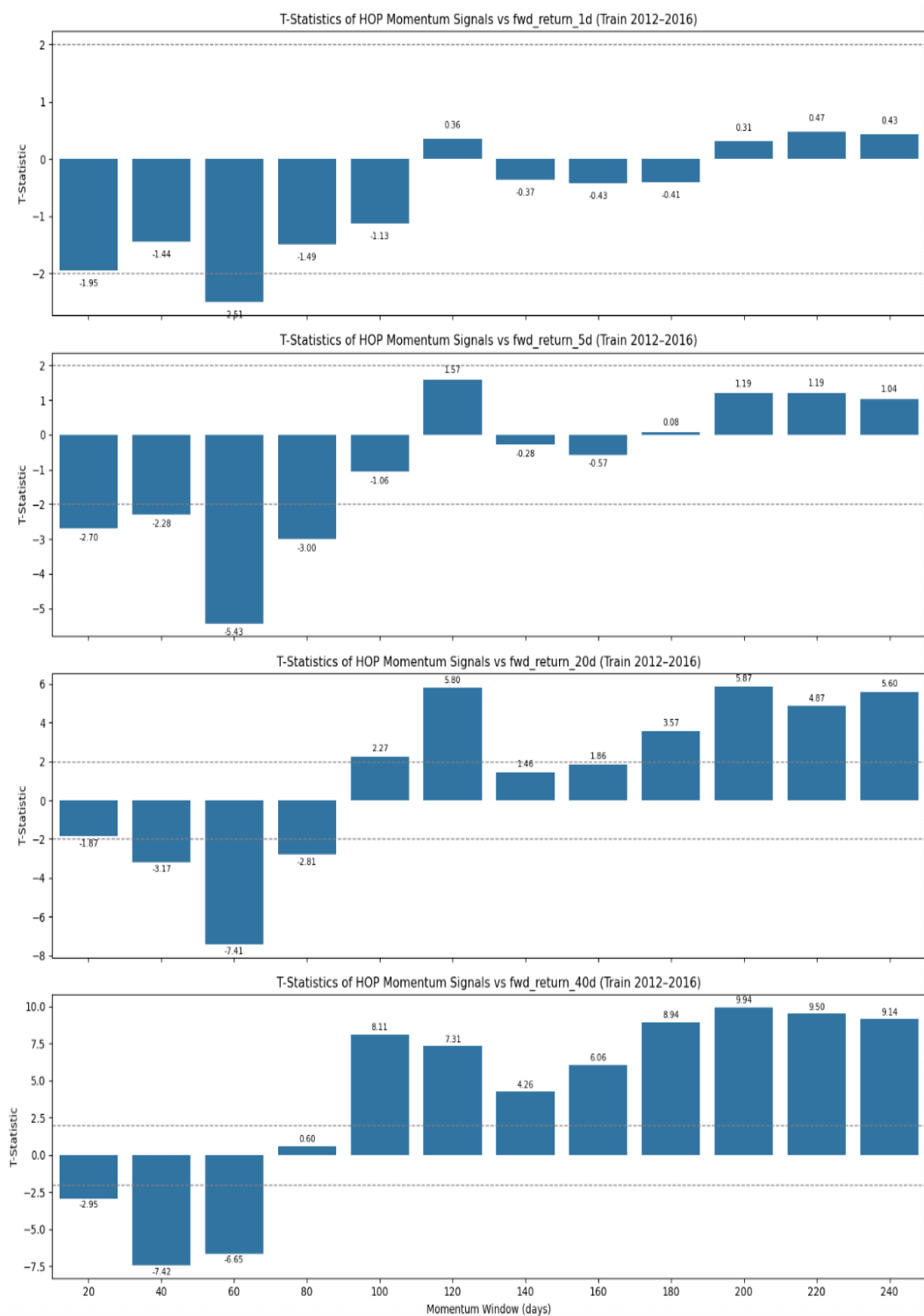


Figure 8 OLS Result of HOP Momentum

f. EWMA, RSI and MACD

In addition, several other classic momentum-related signals were also developed. EWMA Trend and Crossover capture momentum strength through exponentially weighted moving averages over different spans, while RSI (Relative Strength Index) represents a mean-reversion type momentum signal measuring the strength of relative price changes. MACD (Moving Average Convergence Divergence) is a trend-following signal based on the difference between short—and long-term EMAs.

g. Text Data

Title embedding and sentiment:

Firstly, the FinBERT pre-trained model is used to extract two features from financial news headlines. The first one is the CLS vector (CLS Embeddings), a 768-dimensional vector obtained from the model output corresponding to the news title. It represents the news title's overall semantic meaning by mapping the text into a high-dimensional vector space to capture potential semantic features.

The second is Sentiment Scores, the probability distribution of title belonging to the three types of sentiment: negative, neutral and positive.

However, directly using such high-dimensional embeddings as features may lead to the curse of dimensionality. To address this problem and avoid forward-looking bias, the rolling Principal Component Analysis (PCA) was adopted for the embedding vectors (Bouquet, 2017). In this setup, the PCA model can effectively reduce computational complexity by retaining only the most important semantic information (only 10 principal components were kept in this process) from the embeddings.

Intraday news aggregating:

When aggregating the PCA vectors of intraday news for individual stock, a sentiment intensity weighting mechanism was applied.

Specifically, for each piece of news i , the sentiment weight is defined as:

$w_i = \max(\text{sent_neg}_i, \text{sent_neu}_i, \text{sent_pos}_i)$, w_i represents the confidence level of the model in judging the sentiment of this news. The larger the value, the greater influence of this news in the aggregated result.

Subsequently, each PCA principal component k is scaled by weight:

$$\text{PCA}_{k,i}^{\text{weighted}} = w_i \cdot \text{PCA}_{k,i}$$

Finally, on the same stock s , all the news on the same day d weighted average: $PCA_k^{(d,s)} = \frac{\sum_{i \in A_{d,s}} w_i \cdot PCA_{k,i}}{\sum_{i \in A_{d,s}} w_i}$, $A_{d,s}$ represents the set of news corresponding to date d and stock s .

4. Prediction Model Construction

Data splitting

The rolling window is adopted to avoid look-ahead bias. For each test year (2017-2020), the data of the previous five years is used as the training set, and each test year's data is used as the test set.

Machine Learning Mechanism

The model was trained on a combined dataset containing all stocks, instead of training different models for each individual stock separately. This pooled regression allows the model to capture cross-sectional relationships among stocks.

Lasso and XGBoost:

Lasso applies L1 regularization in regression, penalizing some features' coefficients to zero. Therefore, it can be used to select features and avoid overfitting. XGBoost generates trees iteratively with gradient boosting. It is based on the second-order gradient information and regularization which can help to improve predictive accuracy and generalization.

VSN+LSTM Model (Madmoun, 2025):

For the VSN+LSTM model, the Variable Selection Network (VSN) is used to apply weighted fusion of features at each time step, and the importance of each feature is dynamically controlled through the attention mechanism ([Figure 9](#)). Subsequently, the processed feature sequence is input into the Long Short-Term Memory (LSTM) network ([Figure 10](#)). LSTM structure allow the model to receive multi-day sequence data and effectively store and update long-term memory through the gating mechanism, which is highly suitable for processing time series financial data. (Due to the high complexity of the model, only the 40d forward return is predicted here)

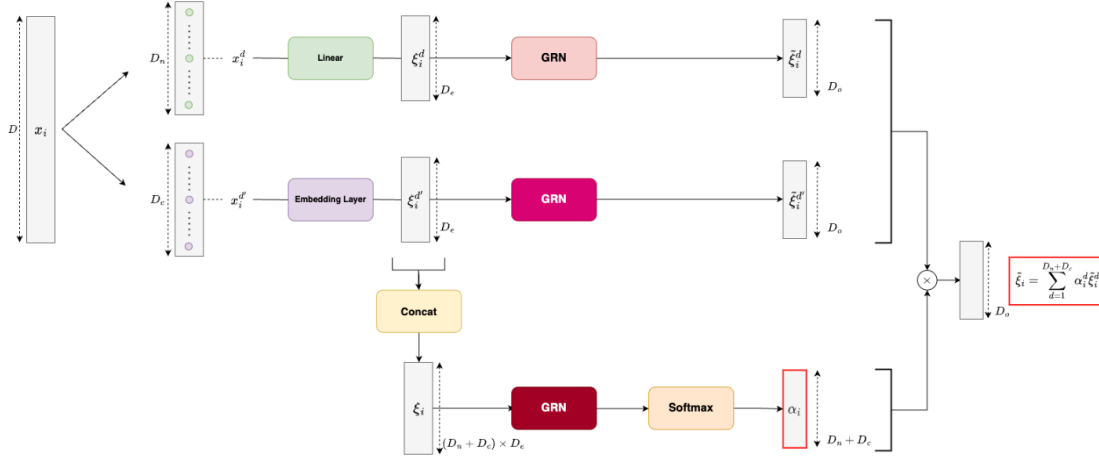


Figure 9 Architecture of the VSN with Attention Mechanism (Madmoun, 2025)

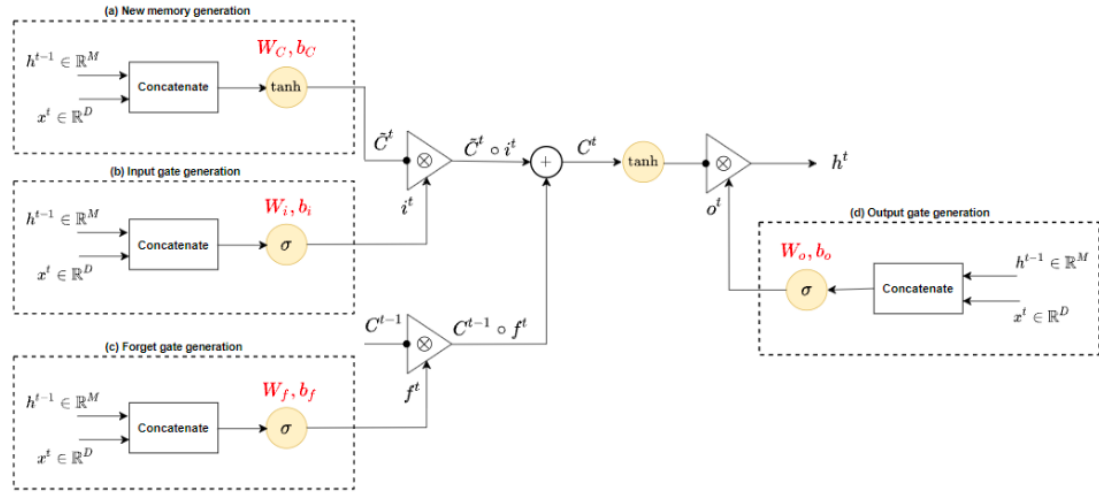


Figure 10 Gating Mechanisms in the LSTM Network (Madmoun, 2025)

Model Performance Evaluation

A. Lasso versus XGBoost (Figure 11):

On the training set, Lasso outperforms XGBoost in both Mean Square Error (MSE, which measures the average square difference between the predicted value and the actual value, smaller is better) and IC (Information Coefficient, which measures the ranking correlation between predicted values and actual values, larger is better). Although the MSE of the 20-day forward return is lower than that of the 40-day one and the IC is only slightly lower, since the subsequent long-short combination pays more attention to the accuracy of the ranking rather than the precision of the predicted value, the 40-day outlook is chosen as the target, and Lasso is used as the main model.

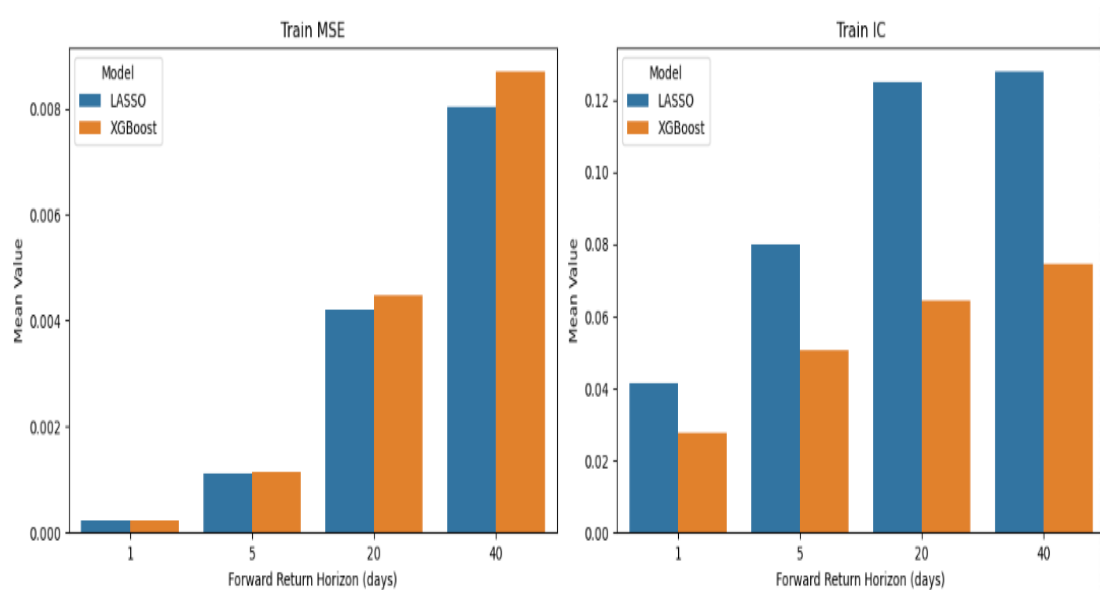


Figure 11 Lasso vs XGBoost in MSE and IC

Lasso Feature Importance Analysis:

Figure 12 shows that Lasso's feature selection in 2017-2018 was diverse, relying on various kinds of signals such as value, momentum signals, and text PCA features. However, only the classic breakout signal still demonstrated a relatively higher feature importance and other signals' coefficients close to zero in 2019-2020, which indicates the Lasso model may failed during this period.

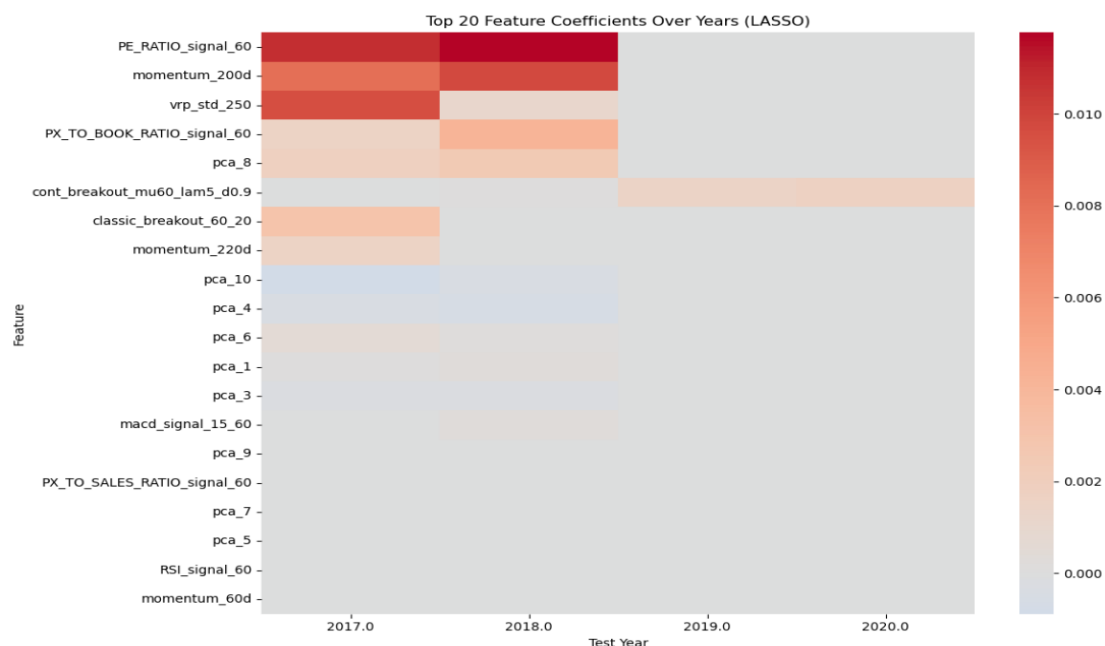


Figure 12 Lasso Time Series Feature Importance

B. VSN+LSTM Model

MSE and IC Performance in Test Set:

Figure 13 shows that in 2017, the MSE was relatively low and the IC score was relatively high, indicating that the model had a good predictive ability during this period. However, not only the MSE rose sharply but also the IC dropped significantly in 2020, indicating that the basic model failed in that period. This phenomenon might be related to the abnormal market regime caused by the COVID-19 pandemic.

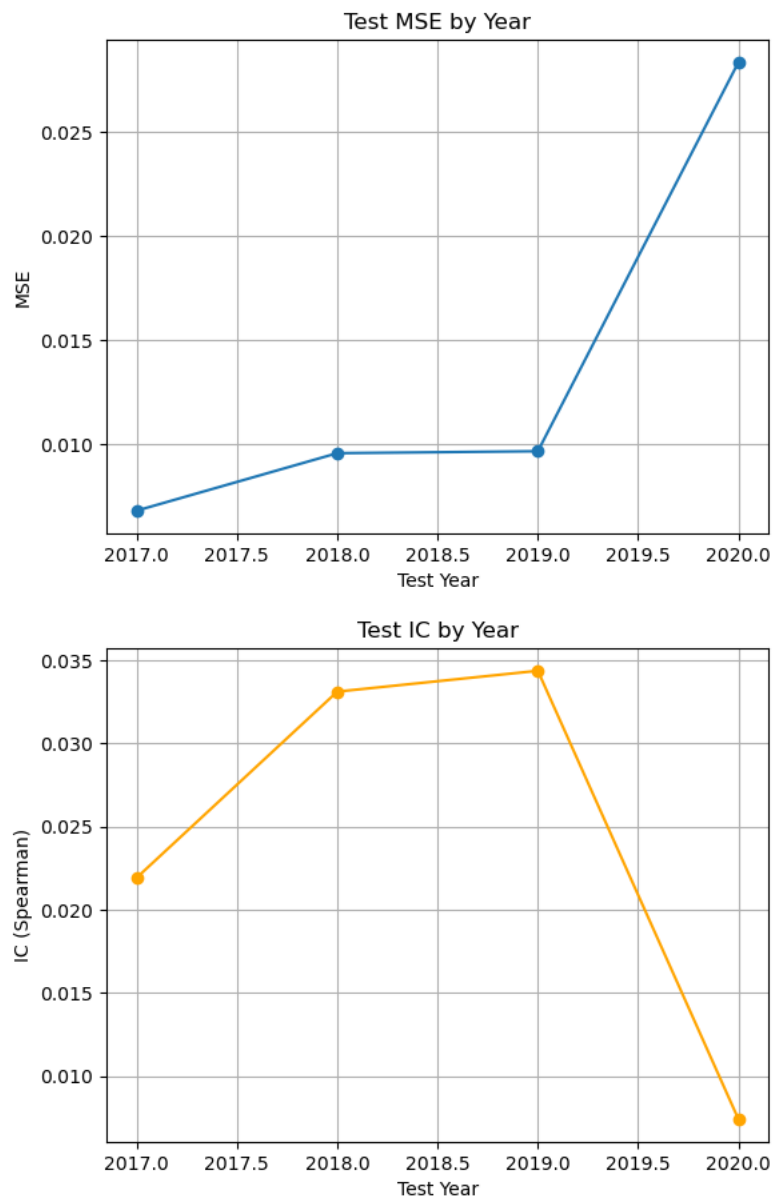


Figure 13 VSN+LSTM MSE and IC Performance in Test Set

Temporal Analysis of VSN Attention Weights:

Figure 14 shows that the VSN+LSTM model in 2017 mainly relied on classic breakout, momentum, and VRP signals. In 2019, it significantly switched to text features and various momentum signals, reflecting that the prediction in that period relied more on trends and news information. In addition, the attention weight of various characteristics tends to be equal in 2020, indicating that the model may lack a stable forecast power in that year.

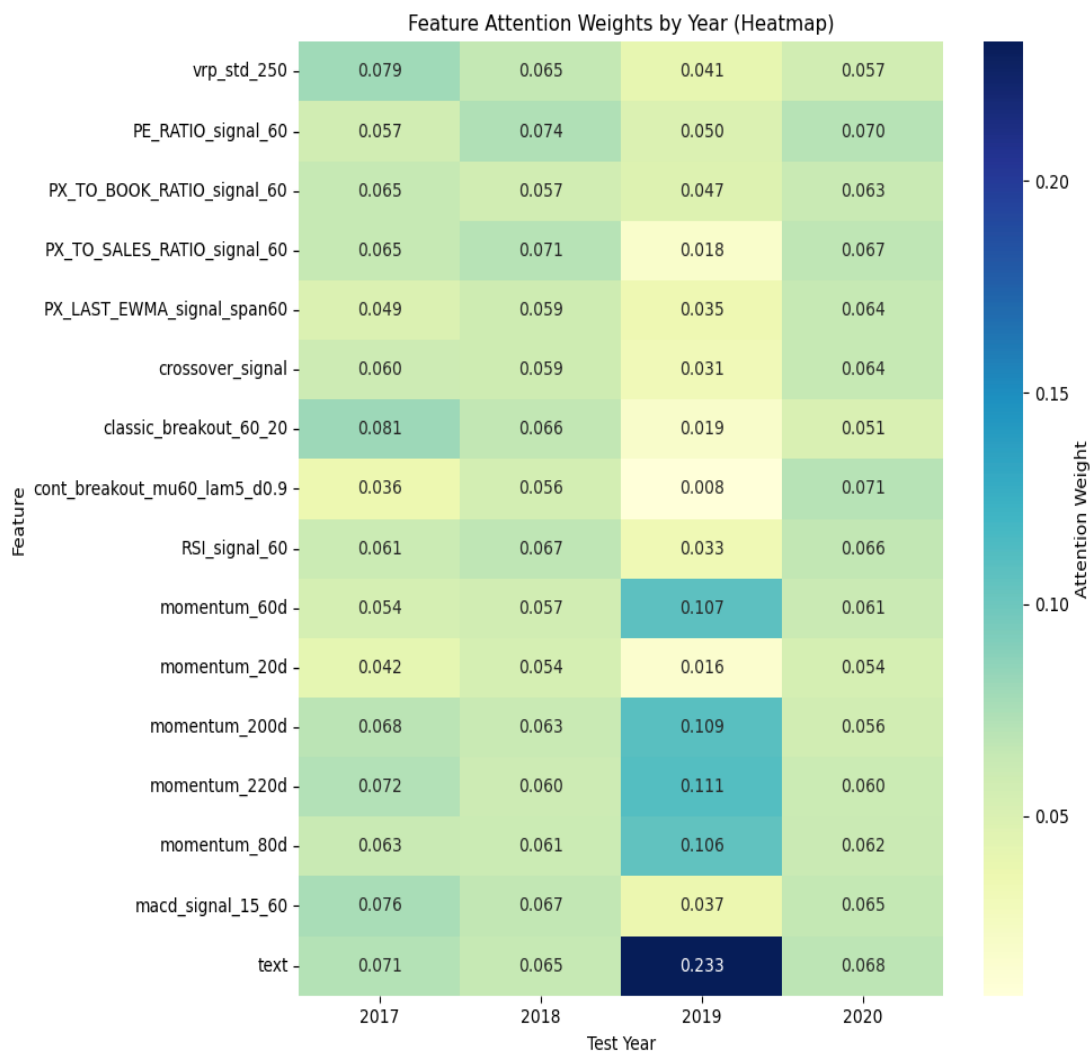


Figure 14 Time Series Feature Attention of VSN

5. Backtest Construction

The backtest was constructed using three performance metrics: Sharpe Ratio, Maximum Drawdown, and Cumulative Return.

Grid Search with Overlapping Long–Short Portfolios:

This part adopts the overlapping mechanism: each position is held for 40 days, and a new position is allowed to be added during the previous position holding period (the interval is the overlapping period, as the parameter to be adjusted). Under this rule, a joint grid search is performed on the long position number, short position number, and overlapping period to minimize Max Drawdown. Only 2017 (the original test set) was used for parameter tuning to avoid data snooping.

Backtesting Result in Test Set with Best Parameters (Lasso and VSN+LSTM):

The test set results show that both models work badly in backtesting. The main reason is that both models failed together at the same time in 2020, resulting in a significant expansion of the maximum drawdown and a drop in the Sharpe Ratio. Although they both bring positive cumulative returns in the final, their performance dropped significantly during market shocks, indicating considerable room for improvement in the models ([Figures 15 and 16](#)).

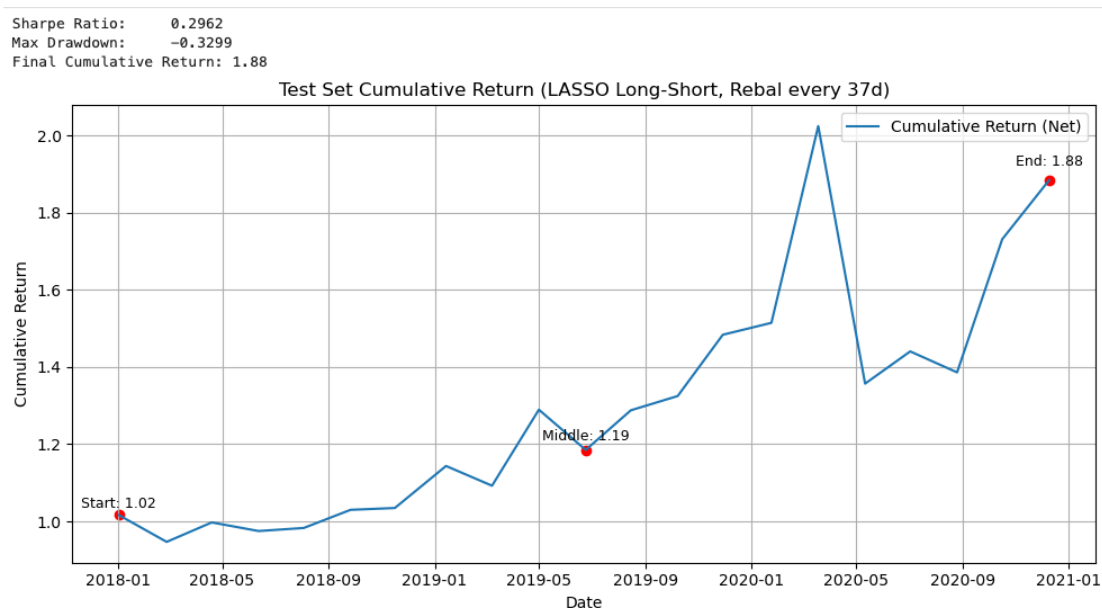


Figure 15 Lasso Backtest Result

Test 2018–2020 Performance with Best Config:

n_long: 4
n_short: 2
rebalance_interval: 37 days
Sharpe Ratio: 0.2105
Max Drawdown: -35.86%
Cumulative Return: 38.18%

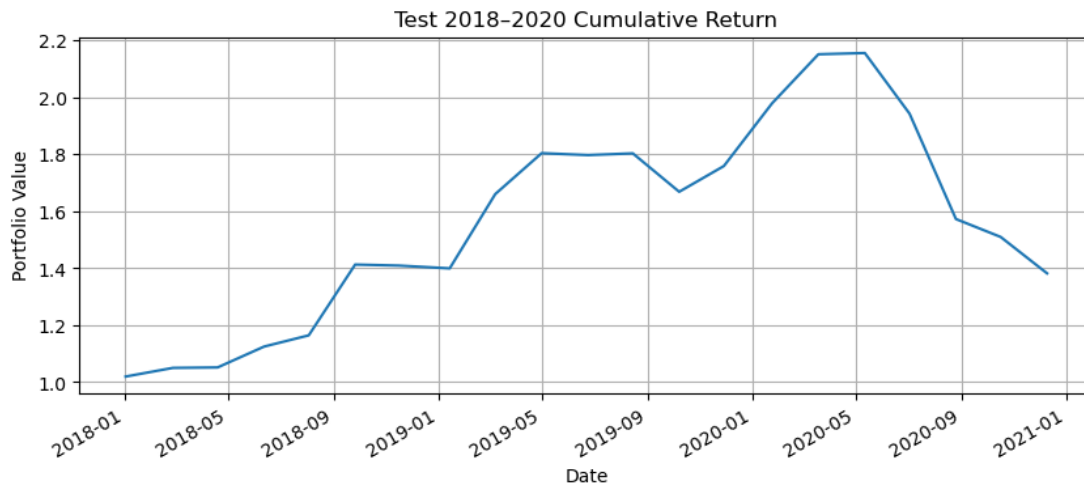


Figure 16 VSN+LSTM Backtest Result

Rolling-Forward Meta Model (Separately for each basic model):

To solve the problem of model failure mentioned above, the meta-model was thus constructed.

The label (Y) is constructed by calculating the daily portfolio return based on the prediction of basic models and the best long-short configuration obtained by Grid Search. If the return is positive, it will be labelled 1; otherwise, 0.

In addition, the meta-models' features are constructed by market regime variables such as the closing price of the S&P 500, the volatility index (VIX), credit spreads (CDX IG and CDX HY), and the risk-free interest rate.

Subsequently, the model adopts the AdaBoost binary classifier to enhance the learning of key patterns by increasing the weights of misclassified samples. This mechanism allows next generated tree to eliminate the mistakes from previous trees.

Ultimately, the predicted labels are used for transaction filtering: no transactions are made when the predicted label is 0, and normal transactions are made when the label is 1.

Basic Model Performance with Meta Model (Lasso and VSN+LSTM):

Figures 17 and **18** show that the meta-model effectively filtered out negative return transactions in both basic models, especially the period during significant market volatility in 2020. Although this process sacrificed some positive return transactions (reduce few true-positive predictions), the maximum drawdown was significantly reduced, and the Sharpe

Ratio was greatly increased (reduce more false-positive predictions). Besides, the cumulative return of the LSTM+VSN strategy has declined, while the cumulative return of the LASSO strategy has increased.

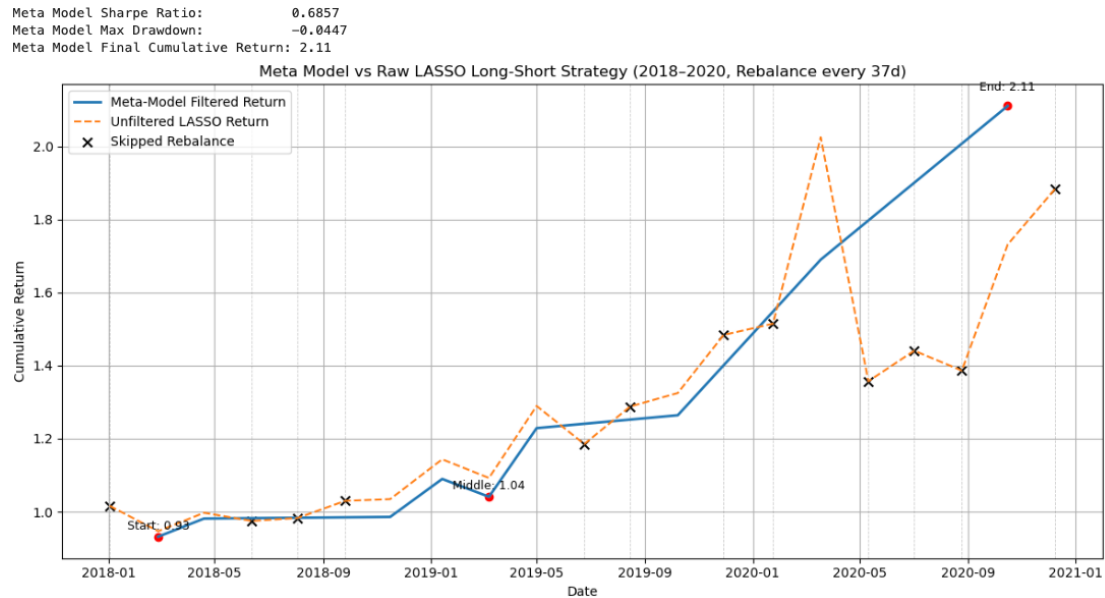


Figure 17 Lasso Backtesting Result with Meta Model

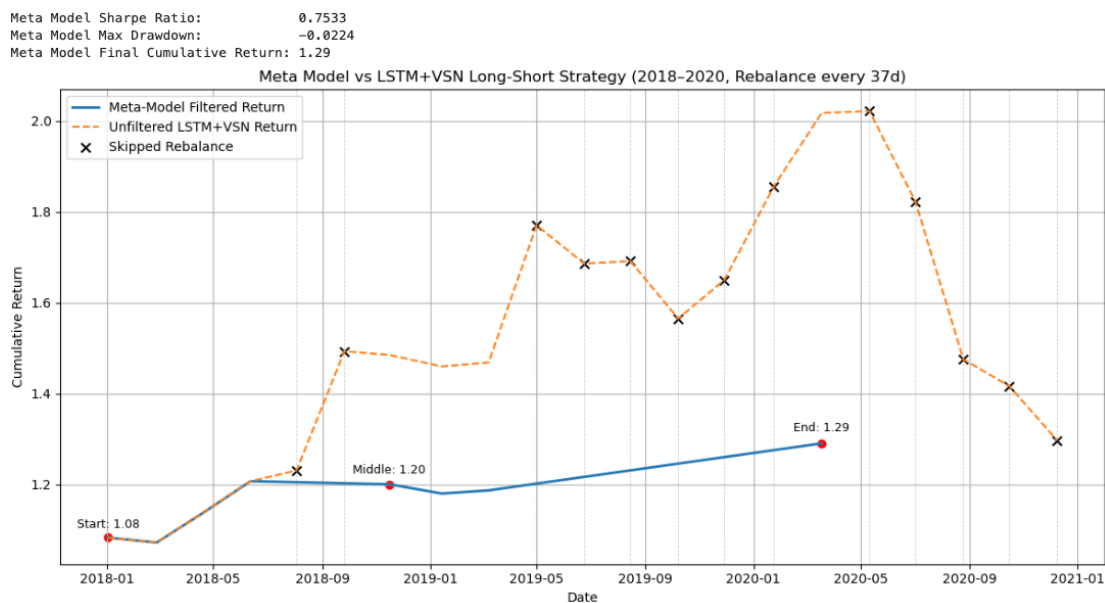


Figure 18 VSN + LSTM Backtesting Result with Meta Model

6. Risk Management (portfolio weights optimization)

Equal Weight: $w_i = \frac{1}{N}$

Volatility Parity: $w_i = \frac{\frac{1}{\sigma_i}}{\sum_{j=1}^N \frac{1}{\sigma_j}}$, Give higher weights to low-volatility assets and reduce the

weights of high-volatility assets to balance risk contributions and minimize extreme drawdowns caused by a single asset.

Risk Parity: $w = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^T \Sigma^{-1}\mathbf{1}}$, Combine asset correlation with volatility to diversify risks and improve the risk-return profile of the investment portfolio.

Optimization Result (Lasso and VSN+LSTM):

Figures 19 and 20 indicate that both volatility and risk parity perform worse than equal weighting. The main reason is that only the equal weighting method was considered in the long-short optimization process. Therefore, adjusting weights after optimization process may break original model and cause worse results. A more reasonable approach is that also using volatility parity and risk parity to calculate individual stock's return and then do long-short optimization process.

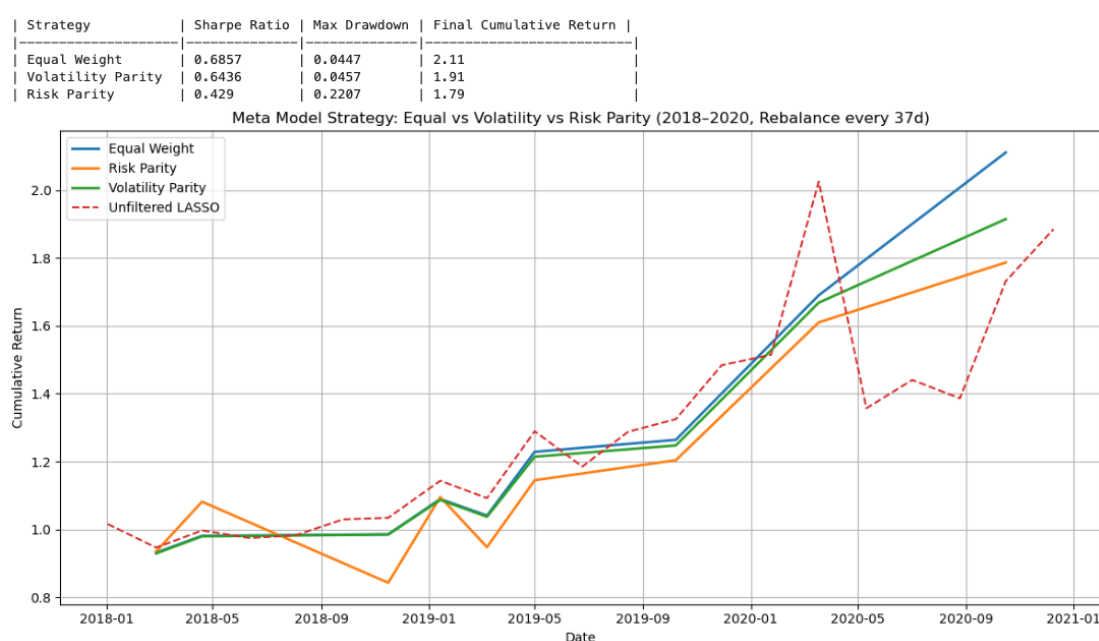


Figure 19 Lasso Portfolio Optimization

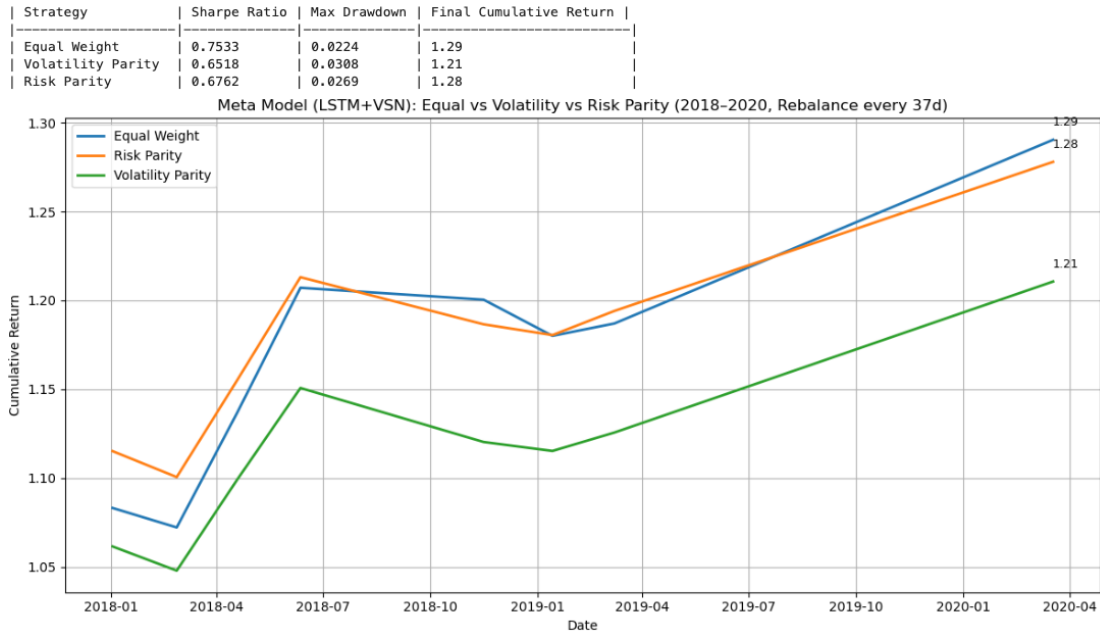


Figure 20 VSN+LSTM Portfolio Optimization

7. Conclusion

This project indicates that the LASSO model's performance is better than VSN+LSTM model's in stock return prediction target. Although LASSO performance is only slightly worse than VSN+LSTM in terms of Sharpe Ratio and maximum drawdown, its final cumulative return is significantly higher. In addition, Lasso is the only model with a cumulative return (111%) exceeding the benchmark S&P 500 (39.32% cumulative return). The possible reason caused VSN+LSTM model's underperformance is that the amount of daily financial market data is inadequate, insufficient to train adequately for deep neural network models (LSTM+VSN). Therefore, in the future research, using hourly level data may help improve VSN+LSTM performance.

On the other hand, applying the meta-model can effectively improve the accuracy of the basic model in trading signal screening, thus significantly improving the strategy return.

8. Appendix

a. Stock Data Structure

Dataset contains the following information for each stock on each trading day: Market price (PX_HIGH, PX_LOW, PX_OPEN, PX_LAST), trading volume (PX_VOLUME), fundamental indicators (PE_RATIO, PX_TO_BOOK_RATIO, PX_TO_SALES_RATIO, CU_R_MKT_CAP), return data (DAY_TO_DAY_TOT_RETURN_GROSS_DVDS), and option implied volatility indicators (IVOL_DELTA, CALL_IMP_VOL_30D, IVOL_IMPLIED_FORWARD).

b. Text Feature Selection and Data Structure Analysis

The text dataset contains the news Date (Date), stock ID (Stock_symbol), news title (Article_title), news Article (Article), and text summary (Textrank_summary).

The news dataset contains 211,165 news articles covering 46 stocks from 2012 to 2020. According to the figure below, there are about 45% missing values in Textrank_summary, and the title field of each news item is non-empty. The title is finally selected as text input for subsequent natural language modelling and signal extraction to avoid data lacking. (**Figure 21**)

```
Total rows: 211165
Non-empty summaries: 115494
Empty summaries: 95671
Unique non-empty summaries: 109784
Number of unique stocks: 46
Date range: 2012-01-01 00:00:00 → 2020-12-31 00:00:00
Empty title: 0
Non-empty title: 211165
Unique non-empty title: 170470
Number of original rows with empty Textrank_summary: 95671
```

Figure 21 Text data summary

In addition, as shown below, the maximum consecutive missing interval for the news title was 4 days (between May 23, 2014 and May 27, 2014). Although there are some missing dates, the overall news data coverage is dense and the missing interval is short, which belongs to the acceptable range and has a limited impact on time series modelling. (**Figure 22**)

```
Maximum news gap: 4 days
Between 2014-05-23 00:00:00 and 2014-05-27 00:00:00
```

Figure 22 Max News Gap

Further analysis (**Figure 23**) shows that a stock may correspond to multiple news on the same day. To cope with this feature, all news on the same day will be aggregated based on sentiment intensity (such as sentiment probability output by FinBERT) in the subsequent processing stage, to construct stable daily text feature input for factor modelling and trading signal generation.

The following figure is an example:

	Date	Stock_symbol	News_Count
4	2012-01-03	CAT	2
6	2012-01-03	MMM	2
7	2012-01-03	MRK	2
10	2012-01-03	TRV	2
15	2012-01-04	BAX	2
17	2012-01-04	CAT	3
20	2012-01-04	CME	2
21	2012-01-04	COF	2
22	2012-01-04	DUK	2
23	2012-01-04	EBAY	2

Figure 23 Each ticker's news counts each day

c. Missing Value Handling

Firstly, for stocks which price are missing, delete them to ensure the accuracy of return calculations. Subsequently, to comprehensively assess the proportion of missing features, a heat map visualization was conducted on the missing proportion of each stock in various feature dimensions. It indicates that PE_RATIO is missing by approximately 20% in about half of the stocks, PX_TO_BOOK_RATIO is seriously missing in a few stocks, while the missing ratios of IVOL_DELTA and IVOL_IMPLIED_FORWARD are relatively small. (**Figure 24**)

Therefore, stocks with proportion of any missing feature exceeds 20% were directly deleted to avoid the impact to the model. The rolling mean of the previous 5 days is used to fill in the remaining missing values. If the rolling window cannot be constructed, the rollback is filled with 0.

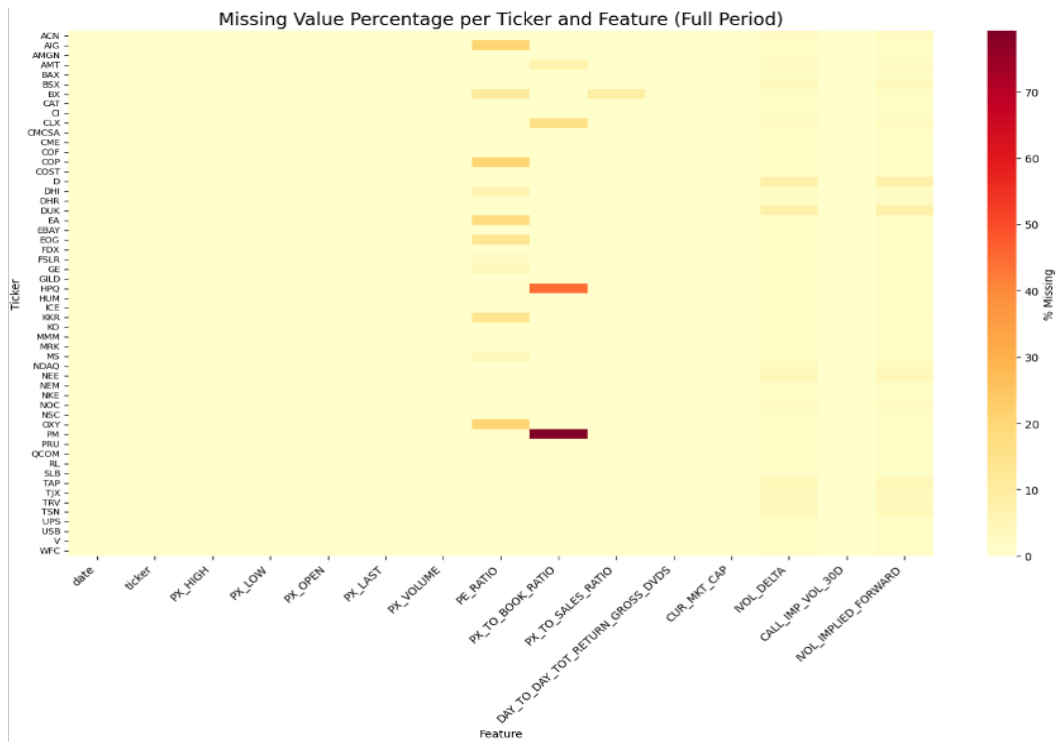


Figure 24 Missing Value Percentage per Ticker and Feature (Full Period)

d. Outlier Handling

During the construction of the carry signal, it is found that the price (PX_LAST) deviates significantly from the implied forward price (IVOL_IMPLIED_FORWARD) in part of the sample (shown below). This abnormal deviation may be caused by the incorrect implied forward price provided by Bloomberg. **(Figure 25)**

To filter these outliers, the forward ratio is introduced: $\text{forward_ratio}_t = \frac{\text{IVOL_IMPLIED_FORWARD}_t}{\text{PX_LAST}_t}$

This ratio measures the degree of difference between the implied forward price and the spot price. In a normal market, it should be close to one. Therefore, if the forward ratio is greater than 1.3 (upper threshold), the implied forward price is considered abnormally high, and if it is less than 0.7 (bottom threshold), it is considered abnormally low.

For observed abnormal implied forward price, setting it equal to the spot price so that the carry signal is neutralized to zero.

Example anomalous observations:					
	ticker	date	PX_LAST	IVOL_IMPLIED_FORWARD	forward_ratio
8307	BAX	2010-01-04	31.6665	58.0509	1.833196
8308	BAX	2010-01-05	31.4492	57.9008	1.841090
8309	BAX	2010-01-06	31.5579	58.3005	1.847414
8310	BAX	2010-01-07	32.0685	59.0008	1.839837
8311	BAX	2010-01-08	32.1446	59.1257	1.839366

Figure 25 Example anomalous observations

e. Code and Data Set

More details, code and data set can be accessed through this link:

https://drive.google.com/drive/folders/1slYxjsxwI0v614aNnJi4T_h61Yyw3Nko?usp=drive_li nk

9. Reference

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